

The hypothesis that 1 is a positive integer is True.

The conclusion that $1^2 \geq 1$ is True since $\boxed{1^2 \geq 1}$.

By defn. of conditional, $P(1)$ is True.

Notice, the truth of the conclusion does not depend on the truth value of the hypothesis.

$\therefore$ This is an example of a trivial proof $\square$

23) Let $P(n)$ be the proposition "If a and b are positive real numbers, then $(a+b)^n \geq a^n + b^n$." Prove that $P(1)$ is true. What kind of proof did you use?

Solution: $P(1)$: If a and b are positive real numbers, then $(a+b)^1 \geq a^1 + b^1$ is True.

$$(a+b)^1 = a+b = a^1 + b^1$$

$\therefore$ The conclusion $\boxed{(a+b)^1 \geq a^1 + b^1}$ is True.

The hypothesis may or may not be True.

By the defn. of conditional, $P(1)$ is True.

Notice, the truth of the conclusion does not depend on the truth value of the hypothesis.

$\therefore$ This is an example of a trivial proof $\square$

24) Show that if you pick three socks from a drawer containing just blue socks and black socks, you must get either a pair of blue socks or a pair of black socks.

Soln: Suppose, for the sake of contradiction, I get no pair of blue socks ~~nor~~ neither any pair of black socks. $\therefore$ We can get at most 2 socks, where one is a blue sock and the other is a black sock. We can also pick 1 sock, which is either blue or black in colour. This is a contradiction to the fact that we have picked 3 socks.

$\therefore$ If three socks are picked from a drawer containing just blue and black socks, we must get either a pair of blue socks or a pair of black socks. $\square$